



Unit 4 · Outcome 3 · Scientific Investigation

Accuracy, Precision & Validity of Measurements

KK04 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Accuracy, Precision & Validity of Measurements

Accuracy, Precision & Validity of Measurements When you generate your own data in an environmental investigation, the markers — and the science — only trust your numbers if you can describe how good they are. This module unpacks the five words VCAA expects you to use precisely: accuracy , precision , repeatability , reproducibility and validity . ⌚ ~22 min read U·O3 · KK04 Key science skills: generate & evaluate 5 practice Qs · 22 marks 3D target loading... (a clustered set of shots on a bullseye) SCENE 1 Four shooters, one target. Watch how where th

MAP The five words, in one breath

MAP The five words, in one breath Before the detail, here's the whole dot point at a glance. Two questions are about a set of measurements (accuracy, precision). Two are about whether other people/conditions get the same answer (repeatability, reproducibility). One is about whether you measured the right thing at all (validity). Accuracy How close a measurement (or their mean) is to the true value . Hurt by systematic error . Precision How close repeated measurements are to each other (their spread). Hurt by random error . Repeatability Same answer

A Accuracy — closeness to the true value

A Accuracy — closeness to the true value VCAA key science skills Accuracy How close a measurement is to the true (accepted) value of the quantity being measured. Accuracy is judged against a reference. You literally cannot say a number is accurate unless you know — or can reasonably accept — what the right answer is. At Mullawa Creek that reference is the calibration standard at 8.20 mg/L . After taking five readings of that standard, a group works out the mean and compares it to 8.20. Group Five readings of the 8.20 standard (mg/L) Mean Accurate? A · fr

Group	Five readings of the 8.20 standard (mg/L)	Mean	Accurate?
A · fresh probe	8.18, 8.22, 8.19, 8.21, 8.20	8.20	Yes — mean $\approx$ true value
B · skipped calibration	8.71, 8.69, 8.72, 8.70, 8.68	8.70	No — mean is +0.50 too high

B Precision — closeness of repeats to each other

B Precision — closeness of repeats to each other VCAA key science skills Precision How close a set of measurement values are to one another . Precision gives no indication of how close those values are to the true value. Precision is about spread , not correctness. A tight cluster of readings is precise; a scattered set is imprecise. The usual culprit behind poor precision is random error — small, unpredictable fluctuations that push readings up and down (probe drift, ripples on the water, you reading the scale a hair differently each time). Group Five r

Group	Five readings of the 8.20 standard (mg/L)	Spread	Precise?
A · fresh probe	8.18, 8.22, 8.19, 8.21, 8.20	± 0.02	Yes — tight cluster
C · ageing probe	7.9, 8.6, 8.0, 8.5, 8.1	± 0.35	No — scattered

LAB The calibration bench — take readings, watch quality emerge

LAB The calibration bench — take readings, watch quality emerge This is the whole dot point in one toy. Pick a scenario, then hit Take reading a few times. Each shot lands on the 3D target and drops a dot on the number line below. The gauges compute the mean (→ accuracy vs the 8.20 true value) and the spread (→ precision) live.

SCENE 2 Mullawa calibration bench — true value 8.20 mg/L drag to orbit · the bullseye = true value Interactive 3D target unavailable on this device — the number line and gauges below still work. Scenario A · calibrated B · offset

⚖ Accuracy vs precision — the distinction examiners love

⚖ Accuracy vs precision — the distinction examiners love If you take one idea from this module, take this: accuracy and precision are independent . A measurement can be any combination of the two. The target picture below is worth memorising — it shows up in some form in almost every set of practice exams. DIAGRAM The four quadrants

Accurate Not accurate Precise Ideal. Tight cluster on the bullseye. Calibrated probe. Dangerous. Tight cluster, wrong spot → systematic error . Looks trustworthy but isn't. Not precise Lucky. Scattered, but the mean lands near true → random error , big averages needed. Useless. Scattered and off-centre → both errors present.

	Accurate	Not accurate
Precise	Ideal. Tight cluster on the bullseye. Calibrated probe.	Dangerous. Tight cluster, wrong spot → systematic error . Looks trustworthy but isn't.
Not precise	Lucky. Scattered, but the mean lands near true → random error , big averages needed.	Useless. Scattered and off-centre → both errors present.

C The two errors behind it all

C The two errors behind it all Accuracy and precision each have a matching error type. Sorting causes into the right bucket is a fast way to lock this in — and it's a common 2–3 mark exam task. Systematic error → hurts accuracy A consistent shift in one direction . Same size every time. Not reduced by repeating. Fix the cause. Random error → hurts precision Unpredictable scatter, both directions . Averages out over many repeats. Reduced by repeating & averaging. Drag each cause into the error it produces. Systematic error Random error Check

D Repeatability — same you, same conditions

D Repeatability — same you, same conditions VCAA key science skills Repeatability The closeness of agreement between successive measurements of the same quantity, carried out under the same conditions of measurement — same method, same instrument, same observer, same location, repeated over a short period . Repeatability is precision under locked-down, identical conditions . At Mullawa Creek: Priya takes five DO readings at the same sampling point, with the same probe, within ten minutes. If those five readings agree closely, her measurement is repeatabl

E Reproducibility — change something, still agree

E Reproducibility — change something, still agree VCAA key science skills Reproducibility The closeness of agreement between measurements of the same quantity, carried out under changed conditions of measurement — e.g. a different observer, different instrument, different method, different location, or different time.

Reproducibility is the tougher, more convincing test. At Mullawa Creek: a second group, using a different probe on a different day, returns to the same point and gets the same DO. If independent conditions still agree, confidence in the results is increased.

What changes?	Repeatability	Reproducibility
Observer	Same	Can differ
Instrument	Same	Can differ
Method	Same	Can differ
Location / time	Same place, short time	Can differ
Tells you...	Your own technique is consistent	The finding holds up independently

F Sort the quality

◇ ACTIVE RECALL

F Sort the quality Each statement below describes one of the five qualities. Drag it to its bin — this is the exact skill behind "identify/justify which quality is being described" questions. Accuracy Precision Repeatability Reproducibility Validity Check

G Validity — did you measure the right thing?

G Validity — did you measure the right thing? VCAA key science skills Validity A measurement or investigation is valid if it actually measures what it claims to / sets out to measure, and the evidence genuinely supports the conclusion drawn. Validity is the big-picture judgement that sits above the other four. You can have accurate, precise, repeatable, reproducible numbers — and still answer the wrong question. Two layers to think about: Valid measurement The instrument and technique measure the intended quantity. A DO probe reads dissolved oxygen — not

📝 Command-word decoder

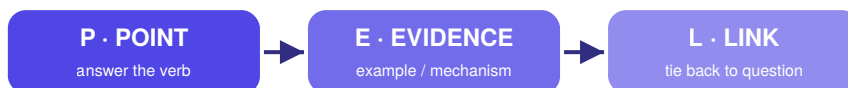
◆ COMMAND WORDS

📝 Command-word decoder The verb in the question tells you how much to write. Tap a card to see what the marker is actually looking for on this dot point.

P·E·L Build a band-6 answer, one move at a time

P·E·L WRITING

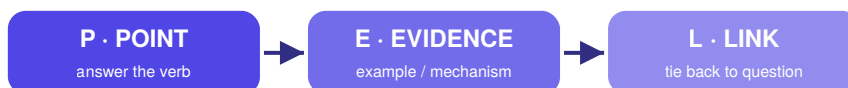
P·E·L Build a band-6 answer, one move at a time Strong science responses follow Point → Evidence → Link . Draft each part below; the keyword meter lights up as you include the language markers worth marks. Then reveal a model to compare.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Worked answer — weak vs strong

 Worked answer — weak vs strong Same question, two responses. The annotations show exactly where marks are won and lost. A student calibrates a DO probe, then records five readings of a standard solution with a known concentration of 8.20 mg/L: 8.71, 8.69, 8.72, 8.70, 8.68 mg/L . Evaluate the accuracy and precision of these measurements. (4 marks)



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice questions — 22 marks

★ PRACTICE & SELF-MARK

Practice questions — 22 marks Attempt each in the box, reveal the marking guide, then self-mark honestly. Your tally feeds the dial in the corner and saves automatically.

★ Key takeaways

✦ RECAP / CHEAT SHEET

★ Key takeaways Lock these in before the exam Accuracy = closeness to the true value ; judged against a reference; ruined by systematic error ; not fixed by repeating. Precision = closeness of repeats to each other (spread); says nothing about correctness; ruined by random error ; improved by repeating & averaging. Accurate and precise are independent — memorise the four-quadrant target. Repeatability = same conditions (same person/gear/place, short time). Reproducibility = changed conditions (different person/gear/method/lab). Validity = measuring the r